Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 2 — Coordinate geometry in the (x, y) plane
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第2章 · 圆的坐标几何
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1.

The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$.

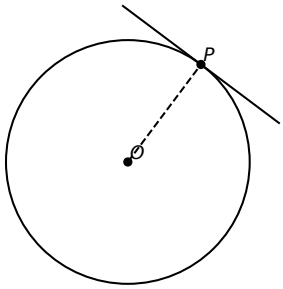


Figure 1

(a) Find its centre and radius. (3)

(b) Find the equation of the tangent at $P = (6, 2)$. (3)

(Total for Question 1 is 6 marks)

$A = (-2, 1)$ and $B = (6, 5)$ are endpoints of a diameter of a circle.

(a) Find an equation of the circle. (3)

(b) Find where the circle meets the line $y = 3$. (2)

(Total for Question 2 is 5 marks)

A circle has centre $(1, 4)$ and passes through $(5, 7)$.

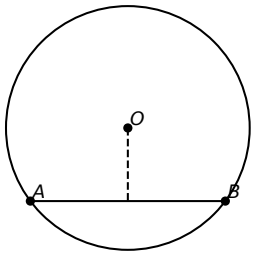


Figure 3

(a) Find its equation. (2)

(b) Find the exact length of the chord cut off by $y = 1$. (3)

(Total for Question 3 is 5 marks)

4.

The circle is $(x+2)^2+(y-1)^2=16$.

(a) Find the intersections with the line $x=0$. (2)

(b) Find equations of both tangents parallel to the y-axis. (2)

(Total for Question 4 is 4 marks)

5.

The line $y = x + k$ is tangent to $x^2 + y^2 = 18$.

(a) Find both possible values of k. (3)

(b) Find the point of contact for $k = 6$. (2)

(Total for Question 5 is 5 marks)

6.

A circle has equation $x^2 + y^2 + 2gx + 2fy = 0$. It passes through $(4, 0)$ and $(0, 6)$.

(a) Find g, f and the centre. (3)

(b) Show that the origin and $(4, 6)$ are opposite ends of a diameter. (3)

(Total for Question 6 is 6 marks)

The circle C is $(x-3)^2+(y+1)^2=25$, and $P=(9,7)$.

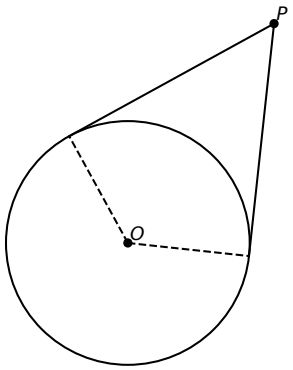


Figure 7

(a) Determine whether P is inside, on or outside C. (2)

(b) Find the exact length of a tangent from P to C. (3)

(Total for Question 7 is 5 marks)

8. P2-C02-R08 · P2 · Chapter 2

A circle has centre $O = (2, -1)$. A chord has endpoints $A = (-1, 3)$ and $B = (5, 3)$.

(a) Find the radius of the circle. (2)

(b) Find the midpoint of AB and verify that the line from O to this midpoint is perpendicular to AB. (3)

(Total for Question 8 is 5 marks)

The circles are $C_1 : x^2 + y^2 = 25$ and $C_2 : (x - 6)^2 + y^2 = 13$.

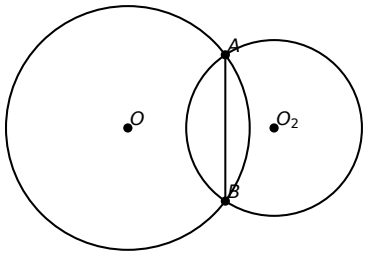


Figure 9

(a) Show that their common chord lies on $x = 4$. (2)

(b) Find the intersection points and the length of the common chord. (3)

(Total for Question 9 is 5 marks)

C is $x^2 + y^2 - 8x - 6y + 16 = 0$.

(a) Find its centre and radius. (3)

(b) Find all tangents to C parallel to the x-axis, including their contact points. (3)

(Total for Question 10 is 6 marks)

The circle $(x - a)^2 + y^2 = 25$ passes through $(1, 3)$. It is given that $a > 0$.

(a) Find a. (3)

(b) Find the equation of the tangent at $(1, 3)$. (3)

(Total for Question 11 is 6 marks)

The line $y = 2$ cuts the circle $x^2 + y^2 = 20$ at A, B.

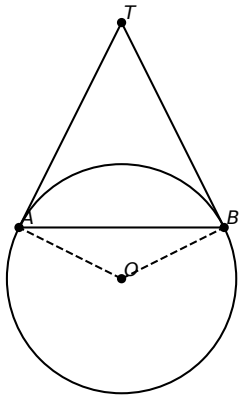


Figure 12

(a) Find A, B and the length AB. (3)

(b) Tangents at A, B meet at T. Find T. (3)

(Total for Question 12 is 6 marks)

13.

A circle has centre $(h, 2)$ and is tangent to the y-axis. It passes through $(4, 4)$, and $h > 0$.

(a) Find h and its radius. (3)

(b) Find its other intersection with the horizontal line $y = 4$. (2)

(Total for Question 13 is 5 marks)

14.

The circle C has centre $(2, 3)$ and is tangent to $y = 7$.

(a) Find an equation of C. (2)

(b) Find the exact coordinates where C meets $y = x + 1$. (3)

(Total for Question 14 is 5 marks)

15.

A circle has diameter endpoints $A = (1, 2)$ and $B = (7, 10)$.

(a) Find its equation and determine whether $P = (0, 6)$ lies on it. (3)

(b) Find an equation of the tangent at A. (3)

(Total for Question 15 is 6 marks)

16.

The circle C is $(x+1)^2+(y-2)^2=10$. The line L is $y=x+4$.

(a) Find both points where L meets C. (3)

(b) Find the midpoint of the chord and show that it lies on the perpendicular from the centre to L. (3)

(Total for Question 16 is 6 marks)

17.

A circle with centre $(3, -2)$ is tangent to the line $x = 8$.

(a) Find its equation and the point of contact. (3)

(b) Find the exact length of the chord on the line $y = 1$. (3)

(Total for Question 17 is 6 marks)

18. P2-C02-R18 · P2 · Chapter 2

The circle is $x^2 + y^2 + 4x - 10y + k = 0$. Its radius is 6.

(a) Find k and the centre. (3)

(b) Determine whether the circle cuts the x-axis. Find any intersections exactly. (3)

(Total for Question 18 is 6 marks)

The line $y = mx$ is tangent to $(x - 5)^2 + y^2 = 9$.

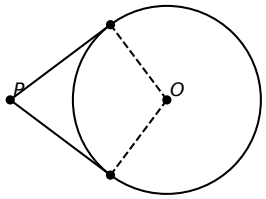


Figure 19

(a) Find the two possible values of m. (3)

(b) Find both contact points. (3)

(Total for Question 19 is 6 marks)

$A = (2, 0)$ and $B = (0, 4)$ lie on a circle whose centre lies on $y = x$.

(a) Find the centre. (3)

(b) Find the circle equation and its intersections with $y = x$. (4)

(Total for Question 20 is 7 marks)

21. P2-C02-R21 · P2 · Chapter 2

The circles $C_1 : (x-1)^2 + y^2 = 9$ and $C_2 : (x-6)^2 + y^2 = 4$ are given.

(a) Show that the circles touch externally. (3)

(b) Find the contact point and the common tangent there. (3)

(Total for Question 21 is 6 marks)

Circle C has centre $(0, 0)$ and radius 5. Point $P = (4, 3)$ lies on it.

(a) Find the tangent L at P. (3)

(b) Find the area of the triangle enclosed by L and the coordinate axes. (3)

(Total for Question 22 is 6 marks)

The circle C is $(x-2)^2+(y+3)^2=13$.

(a) Find the points on C whose x-coordinate is 4. (2)

(b) Find the equations of the tangents at these points and their intersection. (4)

(Total for Question 23 is 6 marks)

A circle has centre $(1, 2)$ and is tangent to the line $y = x + 6$.

(a) Find the foot of the perpendicular from the centre to the line. (3)

(b) Find an equation of the circle. (2)

(Total for Question 24 is 5 marks)

The circle C has diameter endpoints $A = (-1, 2)$ and $B = (5, 10)$. Point $P = (6, 3)$ is given.

(a) Find the equation of C. (3)

(b) Show that P lies on C and that angle APB is 90° using gradients. (3)

(Total for Question 25 is 6 marks)

The circle C is $x^2 + y^2 - 2x - 8y + 8 = 0$. The point $A = (1, 4)$ is given.

(a) Find the centre and radius and explain why A is not on C. (3)

(b) Find all points on C lying on the line $y = 2x + 2$. (3)

(Total for Question 26 is 6 marks)

The circle C is $x^2 + y^2 = 50$. The chord AB has midpoint $M = (3, 1)$.

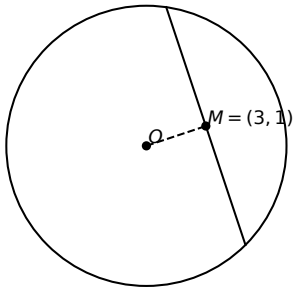


Figure 27

(a) Find an equation of the chord. (3)

(b) Find the exact chord length. (4)

(Total for Question 27 is 7 marks)

28.

A circle has equation $x^2 + y^2 + Dx + Ey + F = 0$ and passes through $(0, 0)$, $(6, 0)$ and $(0, 8)$.

(a) Find D, E, F and the equation in centre-radius form. (4)

(b) Find the tangent at the origin. (3)

(Total for Question 28 is 7 marks)

The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. The line l has equation $y = x + k$.

(a) Find the centre and radius of C. (2)

(b) The line l is tangent to C.

(i) Find both possible values of k. (2)

(ii) For $k = -5 + 5\sqrt{2}$, find the exact coordinates of the point of contact. (2)

(Total for Question 29 is 6 marks)

A circle has diameter AB, where $A(-1, 2)$ and $B(7, 6)$. A horizontal line cuts the circle at P and Q, and $PQ = 6$.

(a) Find an equation for the circle. (2)

(b) Find

(i) the two possible equations of the horizontal line. (2)

(ii) the coordinates of P and Q for the higher line. (1)

(Total for Question 30 is 5 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C02-R01 · 6 marks

(a) · 3 marks

Centre $(3, -2)$, radius 5.

配方得$(x-3)^2+(y+2)^2=25$。

M1	两坐标配方
A1	中心
A1	半径

(b) · 3 marks

$3x + 4y = 26$.

半径向量$(3,4)$，斜率$4/3$，切线$-3/4$，过P整理。

M1	半径梯度
M1	垂直切线过点
A1	方程

Question 2

P2-C02-R02 · 5 marks

(a) · 3 marks

$(x-2)^2+(y-3)^2=20.$

中心为中点(2,3)，半径平方$AB^2/4=(64+16)/4=20$。

M1	中心
M1	半径平方
A1	圆方程

(b) · 2 marks

$(2-2\sqrt{5},3),(2+2\sqrt{5},3).$

线通过中心，代y=3得$(x-2)^2=20$。

M1	正确代入
A1	两交点

Question 3

P2-C02-R03 · 5 marks

(a) · 2 marks

$(x-1)^2+(y-4)^2=25.$

半径平方$4^2+3^2=25$。

M1	两点距离平方
A1	方程

(b) · 3 marks

8.

代$y=1$得$(x-1)^2=16$，端点$x=-3,5$，同高距离8。

M1	联立求两横坐标
M1	取横差
A1	弦长

Question 4

P2-C02-R04 · 4 marks

(a) · 2 marks

$(0, 1 \pm 2\sqrt{3})$.

代x=0后$(y-1)^2=12$。

M1	正确平方关系
A1	两坐标

(b) · 2 marks

$x = -6$ and $x = 2$.

竖直切线位于中心横坐标±半径，即-2 ± 4。

M1	使用圆的水平极端位置
A1	两竖直方程

Question 5

P2-C02-R05 · 5 marks

(a) · 3 marks

$k = \pm 6$.

代入圆得$2x^2+2kx+k^2-18=0$；判别式$4k^2-8(k^2-18)=0$。

M1	代线入圆
M1	判别式等零
A1	两k

(b) · 2 marks

$(-3, 3)$.

相切二次式重根$x=-k/2=-3$，$y=x+6=3$。

M1	求重根
A1	接触点

Question 6

P2-C02-R06 · 6 marks

(a) · 3 marks

$g=-2, f=-3$; centre $(2,3)$.

点(4,0)给16+8g=0；(0,6)给36+12f=0，中心为(−g,−f)。

M1	两点代入
A1	两参数
B1	中心

(b) · 3 marks

Their midpoint is $(2,3)$ and both lie on the circle.

原点由圆式常数0可见在圆上；(4,6)代入16+36−16−36=0；中点为中心。

M1	检查另一点在圆上
M1	计算中点
A1	直径结论

Question 7

P2-C02-R07 · 5 marks

(a) · 2 marks

Outside, since $CP^2 = 6^2 + 8^2 = 100 > 25$.

比较点到中心距离平方与半径平方，无需先开方。

M1	正确距离平方
A1	分类与比较

(b) · 3 marks

$5\sqrt{3}$.

切点处半径垂直切线，勾股长度 $\sqrt{(CP^2 - r^2)} = \sqrt{75}$。

M1	使用直角切线三角形
M1	正确平方差
A1	根式

Question 8

P2-C02-R08 · 5 marks

(a) · 2 marks

5.

OA坐标差$-3,4$，长度5。

M1	半径两点距离
A1	半径

(b) · 3 marks

Midpoint $(2,3)$; AB horizontal and the centre-to-midpoint line is $x=2$, vertical.

先取中点，再比较水平弦和竖直中垂线，不能用无定义的竖线斜率相乘。

M1	中点
M1	两方向
A1	垂直验证

Question 9

P2-C02-R09 · 5 marks

(a) · 2 marks

Subtracting gives $-12x + 36 = -12$, so $x = 4$.

展开第二圆再与第一圆相减，消去x^2+y^2。

M1	展开并相减
A1	公共弦所在直线

(b) · 3 marks

$(4, 3), (4, -3)$; length 6.

$x=4$代第一圆得$y^2=9$，弦竖直用纵差6。

M1	回代解两个y
A1	两交点
A1	弦长

Question 10

P2-C02-R10 · 6 marks

(a) · 3 marks

Centre $(4, 3)$; radius 3.

配方得$(x-4)^2+(y-3)^2=9$。

M1	完成配方
A1	中心
A1	半径

(b) · 3 marks

$y = 0$ at $(4, 0)$ and $y = 6$ at $(4, 6)$.

水平切线在中心正上方和正下方，各相距$r=3$。

M1	正确极端位置
A1	两线
A1	接触点

Question 11

P2-C02-R11 · 6 marks

(a) · 3 marks

$a = 5.$

$(1-a)^2=16$给a=5或−3；正值限制选5。

M1	代点解平方方程
M1	使用正值限制
A1	a

(b) · 3 marks

$4x - 3y = -5.$

半径从(5,0)到(1,3)斜率−3/4，切线4/3；$y-3=(4/3)(x-1)$。

M1	半径斜率
M1	切线负倒数
A1	方程

Question 12

P2-C02-R12 · 6 marks

(a) · 3 marks

$(\pm 4, 2)$; $AB = 8$.

$x^2+4=20$，$x=\pm 4$，弦水平。

M1	联立
A1	两点
A1	长度

(b) · 3 marks

$T = (0, 10)$.

切线为$4x+2y=20$和$-4x+2y=20$，联立$x=0$,$y=10$；这些方程可由半径垂直推得。

M1	两条正确切线
M1	联立
A1	T

Question 13

P2-C02-R13 · 5 marks

(a) · 3 marks

$h = 5/2$; radius $5/2$.

到y轴的距离为h，故r=h；$(4-h)^2+4=h^2$给$20-8h=0$。

M1	切y轴的半径关系
M1	点条件
A1	中心参数和半径

(b) · 2 marks

$(1, 4)$.

关于中心竖直线x=5/2对称，两个交点横坐标和5；另一x=1，或联立求根。

M1	对称或圆线联立
A1	另一点

Question 14

P2-C02-R14 · 5 marks

(a) · 2 marks

$(x-2)^2+(y-3)^2=16$.

中心到水平切线距离7−3=4，即半径。

M1	半径
A1	方程

(b) · 3 marks

$(2\pm2\sqrt{2},3\pm2\sqrt{2})$, with matching signs.

直线过中心，代入得$2(x-2)^2=16$，$y=x+1$。

M1	正确联立简化
M1	求两根
A1	两配对点

Question 15

P2-C02-R15 · 6 marks

(a) · 3 marks

$(x-4)^2+(y-6)^2=25$; P gives $16\neq 25$, so P does not lie on it.

中心(4,6)，r^2=25；给定点(0,6)距中心4并非5。

M1	由直径求圆
M1	检查点
A1	正确结论

(b) · 3 marks

$3x+4y=11$.

半径斜率4/3，切线−3/4，过(1,2)得3x+4y=11。

M1	半径方向
M1	切线过点
A1	方程

Question 16

P2-C02-R16 · 6 marks

(a) · 3 marks

$\left(\frac{-3-\sqrt{19}}{2}, \frac{5-\sqrt{19}}{2}\right)$ and $\left(\frac{-3+\sqrt{19}}{2}, \frac{5+\sqrt{19}}{2}\right)$.

代线入圆得$(x+1)^2+(x+2)^2=10$，即$2x^2+6x-5=0$；二次公式给$x=(-3\pm\sqrt{19})/2$，回代$y=x+4$。

M1	代入建立二次方程
M1	解方程
A1	两点

(b) · 3 marks

Midpoint $(-3/2, 5/2)$, on $y=-x+1$.

二次方程两根平均为$-3/2$，直线给$y=5/2$；中心$(-1,2)$向斜率1直线的垂线为$y=-x+1$。

M1	中点
M1	中心垂线
A1	验证

Question 17

P2-C02-R17 · 6 marks

(a) · 3 marks

$(x-3)^2+(y+2)^2=25$; contact $(8,-2)$.

半径为8−3=5，接触点与中心同纵坐标。

M1	半径
A1	方程
B1	接触点

(b) · 3 marks

8.

距中心竖直3，半弦 $\sqrt{(25-9)}=4$，整弦8。

M1	正确中心距与半弦勾股
M1	从半弦到整弦
A1	长度

Question 18

P2-C02-R18 · 6 marks

(a) · 3 marks

$k = -7$; centre $(-2, 5)$.

配方得$(x+2)^2+(y-5)^2=29-k$；$29-k=36$给$k=-7$。

M1	正确配方
M1	半径条件
A1	k和中心

(b) · 3 marks

It cuts at $(-2 \pm \sqrt{11}, 0)$.

设$y=0$得$(x+2)^2=11>0$，故有两个不同实交点。

M1	联立x轴
M1	判断两实根
A1	两坐标

Question 19

P2-C02-R19 · 6 marks

(a) · 3 marks

$m = \pm 3/4$.

代入得$(1+m^2)x^2-10x+16=0$，判别式$100-64(1+m^2)=0$。

M1	圆线联立
M1	重根条件
A1	两斜率

(b) · 3 marks

$(16/5, 12/5)$ and $(16/5, -12/5)$.

重根$x=10/[2(1+9/16)]=16/5$；$y=m\,x$。

M1	重根横坐标
M1	对应纵坐标
A1	两点

Question 20

P2-C02-R20 · 7 marks

(a) · 3 marks

$(3,3)$.

设中心(h,h)，$OA^2=(h-2)^2+h^2$，$OB^2=h^2+(h-4)^2$；相等得h=3。

M1	中心参数化
M1	两半径平方相等
A1	中心

(b) · 4 marks

$(x-3)^2+(y-3)^2=10$; $(3\pm\sqrt{5},3\pm\sqrt{5})$.

$r^2=1+9=10$；联立y=x得$2(x-3)^2=10$。

M1	半径平方
A1	圆方程
M1	联立过心线
A1	两点

Question 21

P2-C02-R21 · 6 marks

(a) · 3 marks

Centre distance 5 equals the sum of radii $3+2$.

两中心在同一水平线上，距离6−1=5，与半径和相等，所以外切。

M1	中心距与两半径
M1	比较半径和
A1	外切结论

(b) · 3 marks

Contact $(4,0)$; tangent $x=4$.

接触点在连心线上，距第一中心向右3；半径水平所以切线竖直。

M1	定位接触点
A1	点
A1	切线

Question 22

P2-C02-R22 · 6 marks

(a) · 3 marks

$4x + 3y = 25$.

半径斜率3/4，切线−4/3；过(4,3)。

M1	正确垂直斜率
M1	过点
A1	方程

(b) · 3 marks

$625/24$ square units.

截距25/4和25/3，面积为其积的一半。

M1	两截距
M1	正确面积式
A1	精确面积

Question 23

P2-C02-R23 · 6 marks

(a) · 2 marks

$(4, 0)$ and $(4, -6)$.

代x=4得$(y+3)^2=9$。

M1	正确平方式
A1	两点

(b) · 4 marks

$2x + 3y = 8$ and $2x - 3y = 26$; intersection $(17/2, -3)$.

半径向量分别(2,3)、(2,−3)，切线用垂直关系；两线相加4x=34。

M1	第一切线
M1	第二切线
M1	联立
A1	交点

Question 24

P2-C02-R24 · 5 marks

(a) · 3 marks

$(-3/2, 9/2)$.

垂线斜率-1，过中心为y=−x+3，与y=x+6联立。

M1	中心垂线
M1	联立
A1	垂足

(b) · 2 marks

$(x-1)^2+(y-2)^2=25/2$.

半径平方=$(-5/2)^2+(5/2)^2=25/2$。

M1	垂足到中心距离平方
A1	圆方程

Question 25

P2-C02-R25 · 6 marks

(a) · 3 marks

$(x-2)^2+(y-6)^2=25$.

中点(2,6)，半径长度5。

M1	直径中点
M1	半径
A1	方程

(b) · 3 marks

$(6-2)^2+(3-6)^2=25$; $m_{PA}=1/7$, $m_{PB}=-7$, product -1.

先代圆式验证点，再计算过P的两边斜率，乘积−1。

M1	点在圆上验证
M1	两个斜率
A1	垂直结论

Question 26

P2-C02-R26 · 6 marks

(a) · 3 marks

Centre $A = (1, 4)$; radius 3; the centre is not on a circle of positive radius.

配方为$(x-1)^2+(y-4)^2=9$，A恰为中心而非周界点。

M1	配方
A1	中心与半径
B1	排除A在圆上的理由

(b) · 3 marks

$(1 \pm 3/\sqrt{5}, 4 \pm 6/\sqrt{5})$, with matching signs.

直线过中心，用$u=x-1$得$y-4=2u$，圆上$5u^2=9$。

M1	正确联立简化
M1	解正负u
A1	两点

Question 27

P2-C02-R27 · 7 marks

(a) · 3 marks

$3x + y = 10$.

OM斜率1/3，弦垂直OM故斜率−3且过M，y−1=−3(x−3)。

M1	弦与中心连线垂直
M1	用中点过线
A1	方程

(b) · 4 marks

$4\sqrt{10}$.

$OM^2=10$，半弦平方$r^2-OM^2=40$，整弦$2\sqrt{40}=4\sqrt{10}$。

M1	中心距平方
M1	半弦勾股
M1	乘2
A1	最简长度

Question 28

P2-C02-R28 · 7 marks

(a) · 4 marks

$D=-6, E=-8, F=0$; $(x-3)^2+(y-4)^2=25$.

原点给F=0，两个轴点给D=−6,E=−8，再配方。

B1	F
M1	求D和E
M1	配方
A1	标准式

(b) · 3 marks

$3x+4y=0$.

半径从中心(3,4)至原点，斜率4/3；切线−3/4且过原点。

M1	半径方向
M1	正确垂线
A1	方程

Question 29

P2-C02-R29 · 6 marks

(a) · 2 marks

$(3, -2)$; 5.

配方得$(x-3)^2+(y+2)^2=25$。

M1	正确配方
A1	圆心半径

(b) The line l is tangent to C.

(i) · 2 marks

$k=-5\pm5\sqrt{2}$.

代入得$2x^2+(2k-2)x+k^2+4k-12=0$。相切时判别式为零，化简$k^2+10k-25=0$，得$k=-5\pm5\sqrt{2}$。

M1	联立并用判别式
A1	两个参数

(ii) · 2 marks

$(3-5\sqrt{2}/2, -2+5\sqrt{2}/2)$.

接点在斜率-1、过圆心的半径直线$y=-x+1$上。与$y=x-5+5\sqrt{2}$联立。

M1	圆心至切点垂直切线
A1	精确坐标

Question 30

P2-C02-R30 · 5 marks

(a) · 2 marks

$(x-3)^2+(y-4)^2=20$.

圆心是中点$(3,4)$，半径平方$[(8)^2+4^2]/4=20$。

M1	中点与半径
A1	方程

(b) Find

(i) · 2 marks

$y=4\pm\sqrt{11}$.

水平弦中点横坐标为3，半弦为3；若纵坐标为h，则$9+(h-4)^2=20$。

M1	半弦与半径直角关系
A1	两条直线

(ii) · 1 marks

$(0,4+\sqrt{11})$ and $(6,4+\sqrt{11})$.

弦中点$(3,4+\sqrt{11})$，沿水平方向各移动3。

B1	两端点